

Subhankar Roy

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Work Experience

Software Engineer

Apr 2026 – Present

Hake Technologies — Remote, Tripura | Full-time

- Building Angular (TypeScript) frontend features for a GIS web application; consuming FastAPI + PostgreSQL REST APIs and rendering geospatial layers via GeoServer (WMS/WFS).
- Running development integration tests across frontend-backend-GeoServer; identifying CORS, layer-loading, and coordinate-projection issues before production.
- Writing reusable Angular services for WMS/WFS requests; coordinating API contract changes with the backend in Agile sprints.

Software Engineering Intern

Jun 2025 – Jul 2025

National Informatics Centre (NIC), MeitY — Tripura | Internship

- Built 8+ REST endpoints in Spring Boot (Java 17), Hibernate/JPA, and PostgreSQL for Tripura's Professional Tax Portal; handled registration and onboarding workflows.
- Implemented JWT authentication and RBAC; applied parameterised queries and input sanitisation against SQL injection.
- Delivered full-stack features with a React/TypeScript team; resolved cross-origin and data-integrity bugs in a 4-week Agile cycle.

AI Computation Research Intern

Jun 2023 – Jul 2023

Tripura University — Tripura | Research Internship

- Benchmarked CNN training across 4 GPU configurations with PyTorch and CUDA; documented environment setup and performance findings.

AWS Cloud Trainee

Sep 2021 – Jan 2022

F13 Technologies — New Delhi | Trainee

- AWS labs (EC2, S3, RDS, IAM, Lambda, VPC); earned AWS Partner Technical, Business, and Cloud Economics Accreditations.

Projects

Professional Tax Portal (GitHub)

Jun 2025 – Jul 2025

Java 17 · Spring Boot · Hibernate/JPA · PostgreSQL · React · TypeScript · JWT · Docker

- 8+ REST endpoints, JWT auth, RBAC; containerised with Docker; data integrity via DB constraints and JUnit/Mockito tests.

Malware Detection on Linux (GitHub)

Jul 2025 – Jun 2026

Python · scikit-learn · Bash · strace · procfs

- End-to-end detection pipeline using system-call traces (strace/procfs); static and dynamic analysis; ML classification in development.

Single-Stage RISC Processor — ASIC Flow (GitHub)

Feb 2026 – Mar 2026

SystemVerilog · OpenLane · Yosys · OpenROAD

- RTL in SystemVerilog; full RTL-to-GDSII (synthesis → P&R → DRC/LVS) via OpenLane; achieved timing closure at target frequency.

Education

B.Tech, Computer Science and Engineering

Jul 2021 – Jun 2026

NERIST, Arunachal Pradesh Coursework: DSA · OS · Networks · DBMS · Computer Architecture

Certificate Course, Electronics & Communications Engineering

Jul 2019 – Jul 2021

NERIST, Arunachal Pradesh CGPA: 8.4 / 10.0

Technical Skills

Languages Java · Python · TypeScript · C/C++ · SQL · Bash · SystemVerilog

Frameworks FastAPI · Spring Boot · Hibernate/JPA · Angular · React · Maven

Databases PostgreSQL · PostGIS · Redis (familiar)

GIS / Geo GeoServer · WMS/WFS · PostGIS · QGIS (familiar)

DevOps Docker · Git · GitHub Actions (CI/CD) · AWS (EC2, S3, RDS, IAM, Lambda)

Testing JUnit · Mockito · Postman · REST Assured

Security JWT · OAuth 2.0 · RBAC · OWASP Top 10

ML / HW PyTorch · CUDA · RTL design · OpenLane ASIC flow

Certifications

Oracle OCI 2025 Developer Professional (Oct 2025–Oct 2027) | NPTEL: Applied Accelerated AI (Oct 2025) | NPTEL: Deep Learning – IIT Ropar | AWS Cloud Practitioner Essentials (Nov 2021) | AWS Partner Accreditations (2021)